1. Complex numbers

What students need to learn:

Definition of complex numbers in the form $a + \mathrm{i}b$ and $r \cos\ \theta + \mathrm{i}\, r \sin \theta$.	The meaning of conjugate, modulus, argument, real part, imaginary part and equality of complex numbers should be known.
Sum, product and quotient of complex numbers.	$\lvert z_1 z_2 \rvert = \lvert z_1 \rvert\, \lvert z_2 \rvert$ Knowledge of the result $\arg (z_1 z_2) = \arg z_1 + \arg z_2$ is not required.
Geometrical representation of complex numbers in the Argand diagram.	
Geometrical representation of sums, products and quotients of complex numbers.	
Complex solutions of quadratic equations with real coefficients.	
Finding conjugate complex roots and a real root of a cubic equation with integer coefficients.	Knowledge that if z_1 is a root of $\mathrm{f}(z) = 0$ then $z_1{}^*$ is also a root.